Status	Finished
Started	Sunday, 25 October 2026, 12:10 PM
Completed	Sunday, 25 October 2026, 12:15 PM
Duration	5 mins 26 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **2**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the largest element to its correct position
c. Sorting the entire list in one pass
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It repeatedly finds the minimum element and moves it to the sorted part of the list
c. It divides the list into two halves, sorts each half, and then merges them
d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **4**
Complete
Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. Each part is sorted independently
c. The parts are merged without sorting
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **5**
Complete
Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. arrange()
b. sort()
c. order()
d. sorted()

The correct answer is: sort()

Question **6**
Complete
Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Binary Search
b. Bubble Sort
c. Linear Search
d. Merge Sort

The correct answer is: Merge Sort

Question **7**
Complete
Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Provides a basis for other algorithms to work efficiently
c. Decreases the efficiency of algorithms
d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **8**
Complete
Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By breaking the input into smaller parts and merging them
c. By comparing elements sequentially
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **9**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Quick Sort
b. Heap Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **10**
Complete
Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Sorting data in a non-sequential manner
b. Bubbling up the largest element to its correct position with each pass
c. Using the divide-and-conquer approach
d. Being the most efficient sorting algorithm

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **11**
Complete
Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Divide and conquer
b. Greedy method
c. Backtracking
d. Dynamic programming

The correct answer is:
Divide and conquer

Question **12**
Complete
Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It only works with integer arrays
c. It sorts data out of the box efficiently
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **13**
Complete
Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are discarded
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **14**
Complete
Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Linear Search
b. Merge Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **15**
Complete
Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Merging
b. Sorting
c. Searching
d. Complexity

The correct answer is:
Complexity